

ggplot2 Tutorial for Bioinformatics

Zorin Gao bioinfo.gaoi@gmail.com

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ggplot2 Tutorial for Bioinformatics Analysis

RNA-seq ggplot2 ” ” Grammar of Graphics

1. ggplot2 (Layers)

ggplot2

- (Data):
- (Aesthetics):
- (Geometries):
- (Stats):
- (Scales):
- (Facets):
- (Themes):

2.

```
“{r} library(ggplot2)
```

```
ggplot(data = , mapping = aes()) + () + () + () + ()
```

```
## 3.
```

```
### A.      (Aesthetics - aes)
```

```
`aes()``
```

- `x`, `y`:
- `color`:
- `fill`:
- `size`:
- `shape`:
- `alpha`:

B. (Geoms)

(Geom)			
***	----	----	----
`geom_point()`	x, y	PCA (Volcano Plot)	`alpha=0.5` (), `size`
`geom_tile()`	x, y	(Heatmap)	`aes(fill = value)`
`geom_boxplot()`	x, y		`outlier.shape = NA` ()
`geom_violin()`	x, y	Marker Cluster	`trim = FALSE`, `scale = "wid`
`geom_bar()`	x	GO/KEGG	`stat = "identity"` (y)
`geom_density2d()`	x, y	**[Wildcard]**	`color = "blue"`
`geom_jitter()`	x, y		`width = 0.2`, `alpha = 0.3`

4. Practical Examples

Example 1: Basic Scatter Plot (PCA-like)

```
::: {.cell}
```

```
```{r .cell-code}
```

```
Load required library
```

```
library(ggplot2)
```

```
Create sample data similar to PCA results
```

```
set.seed(123)
```

```
pca_data <- data.frame(
```

```
 PC1 = rnorm(100),
```

```
 PC2 = rnorm(100),
```

```
 group = factor(rep(c("Control", "Treatment"), each = 50))
```

```
)
```

```
:::
```